

Total No. of Printed Pages: 1

SUBJECT CODE NO: - RNP-8077
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A & B) (NEP) Sem - I and Sem - II
Examination May/June 2026
Engineering Chemistry

[Time:3:00 Hours]**[Max. Marks: 60]**

N. B

Please check whether you have got the right question paper.

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.no s 2, 3, 4, 5, 6 and 7

Q1 Solve the following questions (any 10)**20**

- a) A 100 mL water sample contains 10 mg/L of $CaCO_3$ and 5 mg/L of CO_2 ions. Calculate the total hardness of the water in mg/L as $CaCO_3$.
- b) Define hardness of water give its unit
- c) Give examples of reinforcement materials
- d) Give reason how $CaSO_4$ is responsible for boiler corrosion
- e) Define coolant give its two examples
- f) is final side product of Fuel cell
- g) $O_2 + 2H_2O + 4e^- \rightarrow$ _____
- h) Differentiate between dry corrosion and wet corrosion
- i) The process of 3D printing requires a digital model of the object, typically in the form of a
 - A) Computer-Aided Design (CAD) file
 - B) Physical blueprint
 - C) Hand-drawn sketch
 - D) Three-dimensional scan
- j) Draw schematic presentation of digital radiography
- k) Why is LNG considered a better alternative fuel?
 - A) Higher emissions
 - B) Higher cost
 - C) Higher calorific value per volume
 - D) Higher ignition temperature

Q2 a) Explain boiler trouble**05**

b) Give properties and applications of CNT

05**Q3 a) Explain proximate analysis of a coal****05**

b) Give Composition properties and applications of CNG

05**Q4 a) Explain computed radiography****05**

b) Writes a note on metallurgical processes

05**Q5 a) 50 ml of water sample required 12.7 ml of 0.02 molar EDTA during titration calculate total hardness of water****05**

b) Give synthesis properties applications of PVC

05**Q6 a) Explain viscosity, viscosity index, cloud point and pour point****05**

b) Explain water as a coolant

05**Q7 a) Write a note on Dry corrosion****05**

b) Explain any sacrificial anodic protection methods

05